

the range of 0.4-0.6A/W in comparison to avalanche photodiodes' 40-80A/W. Thermal sensors have poorer responsivity than photoelectric sensors. For instance, the responsivity of pyroelectric sensors is in the range of 0.5-5 μ A/W.

The shape of the spectral response curve of silicon photodiodes, particularly in blue and UV parts of the spectrum, can be altered by choosing an appropriate manufacturing process. Fig. 1 shows typical spectral response curves of normal, blue-enhanced and UV-enhanced, and low-noise silicon photodiodes. Low-noise spectral response corresponds to photodiodes' photovoltaic mode of operation wherein no external bias is applied to the photodiode. Since dark current is a function of bias voltage magnitude, photovoltaic mode of operation eliminates dark current as a source of noise. In this case, NEP will be lower, thereby allowing greater sensitivity at lower wavelengths but at the cost of slightly lower responsivity at higher wavelengths.

Silicon becomes transparent to radiations of longer than 1100nm wavelength. On the contrary, wavelengths in ultraviolet region are absorbed in the first 100nm thickness of silicon. Even the most careful surface preparation leaves some surface damage, which reduces collection efficiency for this wavelength. High absorption coefficient of silicon in blue- and UV-enhanced spectral region leads to carrier generation within heavily doped p+ (or n+) contact surface of P-N and PIN photodiodes. This causes rapid decrease in quantum efficiency of this region due to shorter lifetime and surface recombination.

Spectral response of photodiodes in blue- and UV-enhanced region is enhanced by minimising near-surface recombination. This is achieved by using techniques such as thin and highly graded p+ (or n+ or metal Schottky) contacts, lateral collection to minimise percentage of heavily-doped surface area and passivation layer. A passivation layer such as silicon nitride, silicon dioxide or

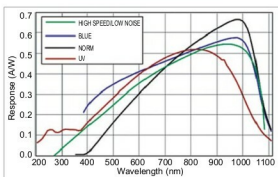


Fig. 1: Spectral response of silicon photodiode

titanium dioxide can reduce surface recombination.

Anti-reflection coatings may be used to enhance responsivity to about 25 per cent at the required wavelength, but this leads to reduced efficiency at other wavelengths that these reflect. Package window also plays an important role in shaping the spectral response. The standard glass window absorbs wavelengths shorter than 300nm. For UV detection, a fused silica or UV transmitting glass window is used. Various filter windows are also available to tailor the spectral response for an application. Optical filters can also be added to change the spectral response. A common example is the use of a specific filter to modify the normal silicon response to approximate the spectral response of the human eye.

Due to improved charge-collection efficiency in photodiodes, responsivity increases slightly with applied reverse bias. It also exhibits dependence on temperature variations due to variation in band-gap energy. The band-gap energy varies inversely with the change in temperature.

The term 'responsivity' should not be confused with 'sensitivity'. Sensitivity is the lowest detectable light level that is typically determined by detection noise. It is also significantly influenced by the required detection bandwidth. Also, a photo-sensor should ideally be operated in a spectral region where its responsivity is not far below the highest possible value, because this leads to the lowest possible detection noise and thus to a high signal-to-noise ratio and high sensitivity.

Noise equivalent power (NEP) is the input power to a sensor that generates an output signal current equal to the total internal noise current of the device, which implies signal-to-noise ratio of 1 (one). In other words, it is the minimum detectable radiation level of the sensor. Noise power and thus noise-equivalent power depend on the assumed detection bandwidth.

If one were to use the full detection bandwidth of a device to compute NEP, the NEP would not allow a fair comparison of sensors with different bandwidths. Therefore it is a common practice to assume a bandwidth of 1Hz, which is usually far below the detection bandwidth. NEP is usually specified in units of W/ $\sqrt$ Hz rather than watts.

Detectivity of a sensor is the reciprocal of its NEP. Sensors with higher value of detectivity are more sensitive. Detectivity, like NEP, depends upon noise bandwidth and sensor area. To eliminate these factors, a normalised figure of detectivity, referred to as D^* (Dee-star), is used. It is defined as the detectivity normalised to an area of 1cm² and noise bandwidth of 1Hz.

Quantum efficiency is the ratio of the number of photoelectrons released to the number of incident light photons absorbed. It is the percentage of input radiation power converted into photo current. It is expressed as rise/fall time parameter in photoelectric sensors and as time constant parameter in thermal sensors. Rise and fall times are the time durations for the output to change from 10 per cent to 90 per cent, and 90 per cent to 10 per cent of the final response, respectively. Rise/fall time parameter determines the highest signal frequency to which a sensor can respond. Time constant is defined as the time required by the output to reach 63 per cent of the final response from zero initial value.

Noise is the most critical factor in designing sensitive radiation detection systems. Noise in these systems is generated in photosensors, radiation sources and post-detection